Corollary: If a is a generator of a finite cyclic group G of order then the other generators of G are the elements of the form a^r wher r is relatively prime to n.

Example: How do all subgroups of $\mathbb{Z}_{18}$ look like.

By the previous theorem $\{1,5,7,11,13,17\}$ generate $\mathbb{Z}_{18}$.

So we want to look at those that $\gcd(k,18) \neq 1$

let's start with $\langle 2 \rangle = \{0,2,4,6,8,10,12,14,16\}$

$\gcd(2,18)=2$ has $\frac{18}{2}$ - elements

~~The~~ which are the other generators of $\langle 2 \rangle$

well: they are of the form $2k$ where k is relatively prime to 9. i.e.

$\{1,2,4,5,7,8\}$ ~~and~~ so the generators

are $\{2,4,8,10,14,16\}$

Ex the element 6 of $\langle 2 \rangle$ generates $\{0,6,12\}$

and 12 is also a generator of this subgroup.

We have found so far all the subgroups generated

by 0, 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 16, 17.

which leaves us to consider only 3, 9, 15

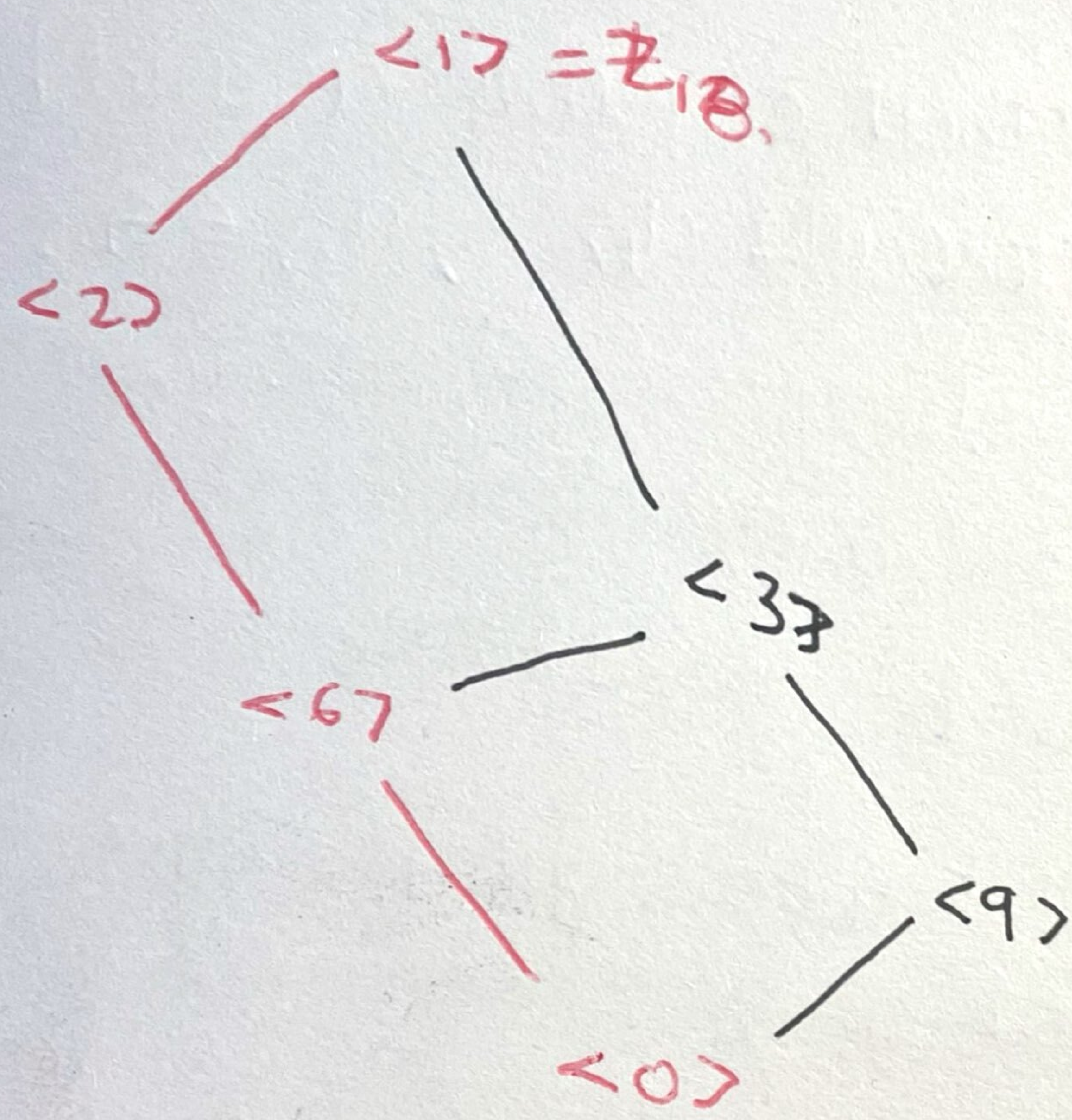
$\langle 3 \rangle = \{0,3,6,9,12,15\}$ order $\frac{18}{3} = 6.$

$\gcd(18,3) = 3$

and 15 generates the same subgroup since

$5 \cdot 3 = 15$ and $\gcd(5,6) = 1.$

Finally, $\langle 9 \rangle = \{0, 9\} \to$ which is the group of order 2.



⟨1⟩ = ℤ18
⟨2⟩
⟨3⟩
⟨6⟩
⟨9⟩
⟨0⟩

Lecture 10: Generating Sets and Cayley Digraphs.

Motivation: We have seen given a G group how to construct the smallest subgroup that contains $a \in G$ which is $\langle a \rangle$.

Goal: What if we want to find the smallest subgroup that contains a and b? or more generally $a_1, \dots, a_n \in G$?

How do the elements would look like?
Given a and b expression of the form

$$a^5 \cdot b^{-3} a^2 b^2$$ should belong to the group

Definition: Let G be a group $(G, *, e)$ and $a, b \in G$ the group

$H =$ of finite expressions of the form

$$\cancel{a^{z_1} b_1^{k_2} \cdots a^{k_n} b}$$

$$\begin{cases} a^{k_1} b_1^{l_1} \cdots a^{k_m} b^{l_m} & \text{for } k_i, l_i \in \mathbb{Z}. \\ b_1^{l_1} a^{k_1} \cdots b^{l_m} a^{k_m} & \end{cases}$$

forms a subgroup and this is the group generated by a and b.

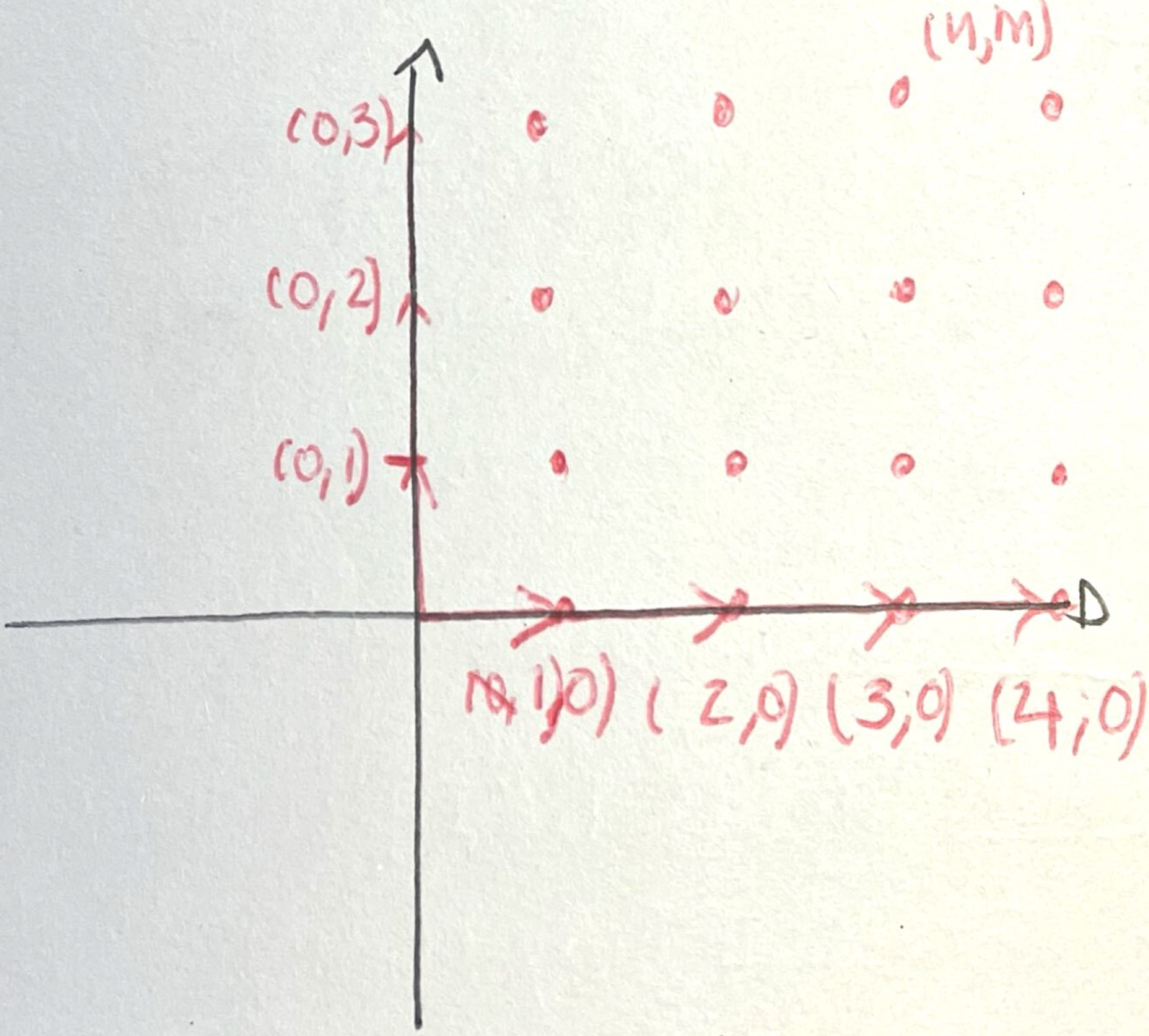
in such case we say that a, b are the generators of H. and write $H = \langle a, b \rangle$.

$\boxed{\text{Example}}$ consider $(\mathbb{R}^2, +, 0)$ where

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}$$

and take $a = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

How does the group generated by it look like? $H = \langle (1,0), (0,1) \rangle = \mathbb{Z}^2$.



We have given an intuitive description of the subgroup of G generated by a, b.

More generally, given $h_1, \dots, h_k \in G$

The group generated by $h_1, \dots, h_k$ consists of all possible finite expressions of elements of powers of the $h_i^{z_i}$ where $z_i \in \mathbb{Z}$.

then $H = \langle h_1, \dots, h_k \rangle$ is the group generated by $\underbrace{h_1, \dots, h_k}_{\text{are the generators}}$ and we say that H is finitely generated.

$\boxed{\text{Then}}$ H is a subgroup of G.

Q:
Is there any other way to construct the smallest subgroup containing $h_1, \dots, h_k$?
↳ Yes!

2nd-approach: As intersections of subgroups.

Definition: Let $\{S_i \mid i \in I\}$ be the collection of sets, here I may be any set of indices. The intersection $\bigcap_{i \in I} S_i$ of sets S_i is the set of elements that are all in the sets S_i
i.e.

$$\bigcap_{i \in I} S_i = \{x \mid x \in S_i \ \forall i \in I\}$$

• If I is finite i.e. $I = \{1, 2, \dots, n\}$ we denote $\bigcap_{i=1}^{n} S_i$ to express $S_1 \cap S_2 \cap \dots \cap S_n$.

Theorem1: Let $(G, *, e)$ be a group and $(H_i)_{i \in I}$ be a family of subgroups. Then $H = \bigcap_{i \in I} H_i$ is a subgroup

Proof:

1) e is in H: since $\forall i \in H_i$ $H_i \leq G$ then $e \in H_i$

Thus $e \in \bigcap_{i \in I} H_i$

2) closed under ~~sum~~ $*$: let $x, y \in H$ then $x, y \in H_i$ $\forall i \in I$

Then since $H_i \leq H$ we have $x * y \in H_i$ $\forall i \in I$

so $x * y \in \bigcap_{i \in I} H_i$.

3) closed under inverse: let $x \in H$ then $x \in H_i$ $\forall i \in I$

then since $H_i \leq H$ $\forall i \in I$. $x^{-1} \in H_i$ $\forall i \in I$ so $x^{-1} \in \bigcap_{i \in I} H_i$ as required.

So given $(G, *, e)$ a group we might take $a_1, \dots, a_n \in G$ then there is at least one subgroup that contains $a_1, \dots, a_n$ which is G itself.

thus if we consider the family of subgroups S_i that contain all the $a_1, \dots, a_n$ the previous theorem tells us that $\bigcap_{i \in I} S_i$ is a subgroup of G.

↳ and it is the smallest subgroup of G, that contains all the $a_1, \dots, a_n$.

$\boxed{\text{Remark}}$ Note that we did not need the a_i's to be finite i.e. we could have taken a family $\underbrace{(a_j)_{j \in I}}_{\text{which is infinite}}$ ~~of infini~~ of elements of G.

<u>Definition</u>: let G be a group and let the $a_i \in G$ for $i \in I$. The smallest subgroup of G containing $\{a_i \mid i \in I\}$ is the subgroup generated by $\{a_i \mid i \in I\}$.
If this subgroup is all of G then we say that $\{a_i \mid i \in I\}$ <u>generates</u> G and the a_i's are the <u>generators</u> of G.
If there is a finite set $\{a_1, \dots, a_n\}$ that generates G we say that G is finitely gen.

Examples:

- Finitely generated $(\mathbb{Z}^2, +, 0)$ where

$$\begin{bmatrix}x_1\\x_2\end{bmatrix}+\begin{bmatrix}y_1\\y_2\end{bmatrix}=\begin{bmatrix}x_1+y_1\\x_2+y_2\end{bmatrix}$$

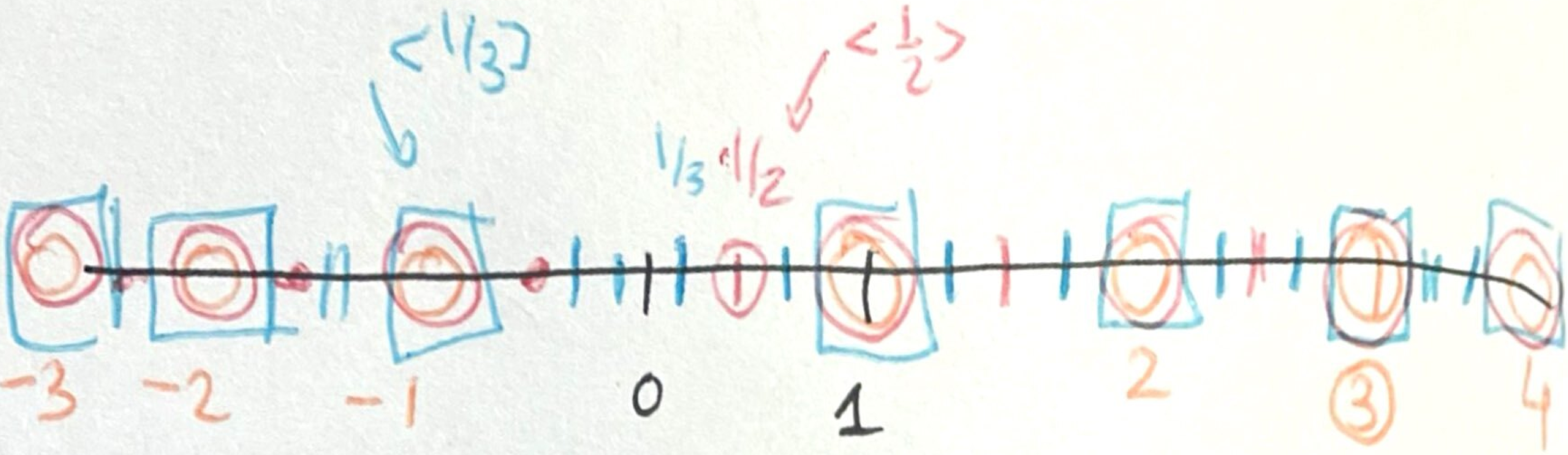
$\to \langle (1,0), (0,1) \rangle$

Picture:

↳ Note it could have been generated by $\langle (1,0), (1,1) \rangle$ since $\begin{pmatrix}0\\1\end{pmatrix}=\begin{pmatrix}1\\1\end{pmatrix}-\begin{pmatrix}1\\0\end{pmatrix}$

In general $(\mathbb{Z}^n, +, 0)$ has n generators.

Not finitely generated $(\mathbb{R}, +, 0)$



$\langle 1 \rangle = \mathbb{Z}$.

in general $\langle \frac{1}{n} \rangle = \frac{1}{n}\mathbb{Z}$.

has a subgroup $\langle \frac{1}{n} \mid n \in \mathbb{N} \rangle$ → not generated by a finite set.

<u>Theorem</u>: If G is a group and $a_i \in G$ for $i \in I$ then the subgroup of H generated by $\{a_i \mid i \in I\}$ has as elements precisely those elements of G that are finite products of integral powers of the a_i, where powers of a fixed a_i may occur several times in the product.

<u>Proof</u>: let K be the set of all finite products of integral powers of the a_i. Then ~~K=H~~. It is just necessary to show that K is a subgroup, since H is the smallest subgroup

containing all the a_i's.

- Note that a product of elements in ~~A~~ K is again in ~~A~~ K.
- $e \in$ ~~A~~ K since $e = a_i^0$
- given $k \in K$ if we form the product giving by taking the order of the a_i's rerenversed and exponent the ~~opp~~ additive opposite of the a_i appearing in k $\Rightarrow$ the inverse gives ~~or~~ something of the required form.

e.g. ~~a_1^3~~ $a_1^3 a_2^{-2} a_3$

we do $a_3^{-1} a_2^2 a_1^{-3}$ ← is the inverse.

Comment: I think the book makes things hard to follow.

1) On one side we have $K = \langle a_i \rangle_{i \in I}$ ← the group generated by the a_i's which are just formal expressions.

2) the point is if we define H to be the smallest subgroup of G containing all the $\langle a_i | i \in I \rangle$

then: $H = K$ so you can think on each of them in literally the same way. The last theorem shows this since $K \supseteq H$ and is a subgroup and contains all the a_i's.

Then if K is the subgroup of formal expressions and $H = \bigcap_{i \in I} S_i$ where S_i are the subgroups containing the a_i's then

$K = S_i$ for some i so $H \leq K$.

But on the other hand the subgroup of formal expression must be contained in any subgroup S_i containing all the $(a_i \mid i \in I)$

then $K \leq S_i \quad \forall i \in I$

so $K \subseteq H = \bigcap_{i \in I} S_i$.

Cayley Graphs:

Main idea: Given a finite group G for each generating set S we will like to associate a graph. (directed)

vertices = elements of G.

generators = edges.

Baby example: say $S=\{a,b,c\}$

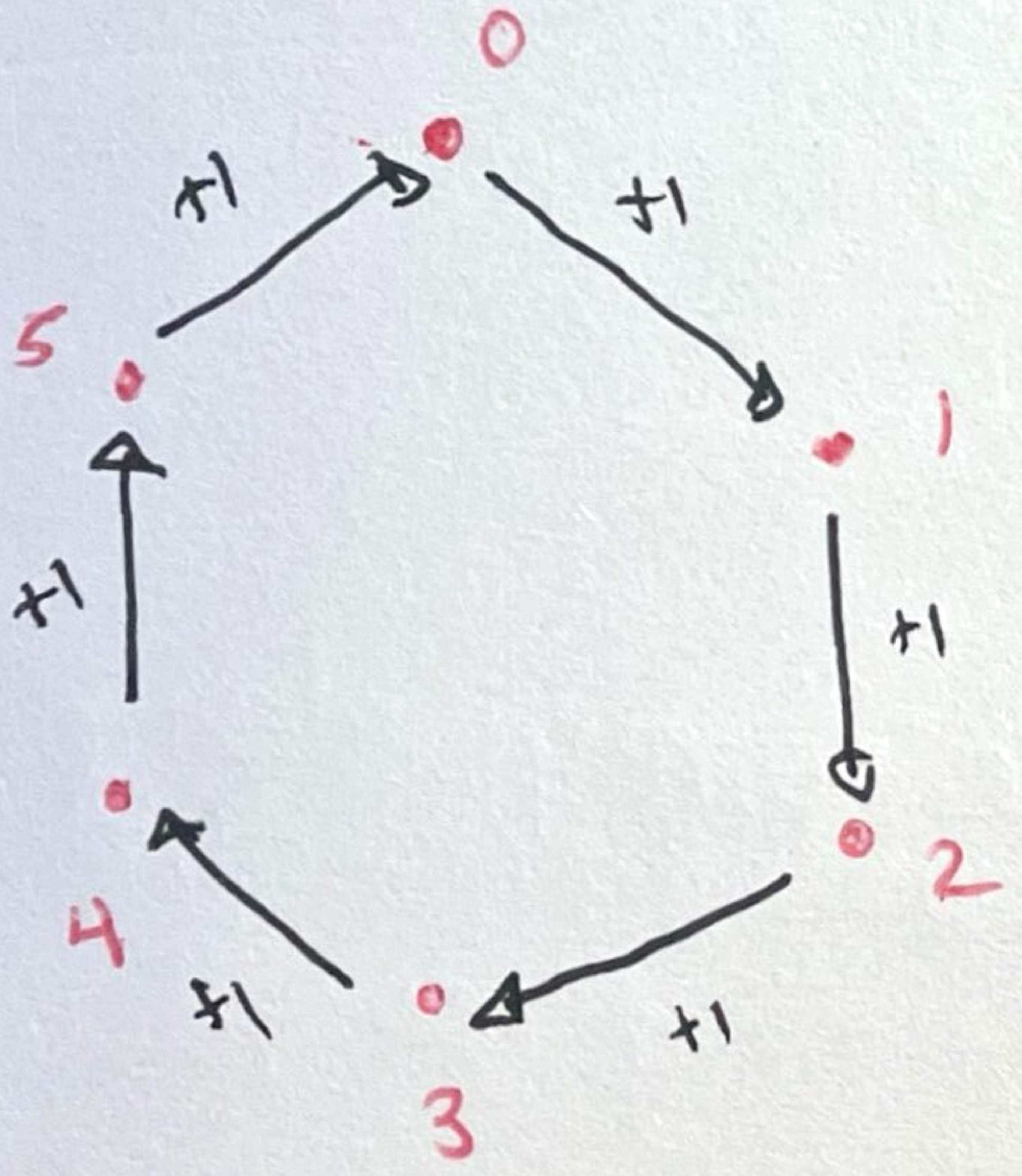
blue red orange

$x \longrightarrow y$ means $y = x \cdot a$.

Since we are in a group $x = a^{-1}y$ thus travelling across the arc in the opposite direction gives us multiplication by a^{-1}!

If a generator b is its own inverse then we omit the direction of the arc i.e. $b \cdot b = e$.

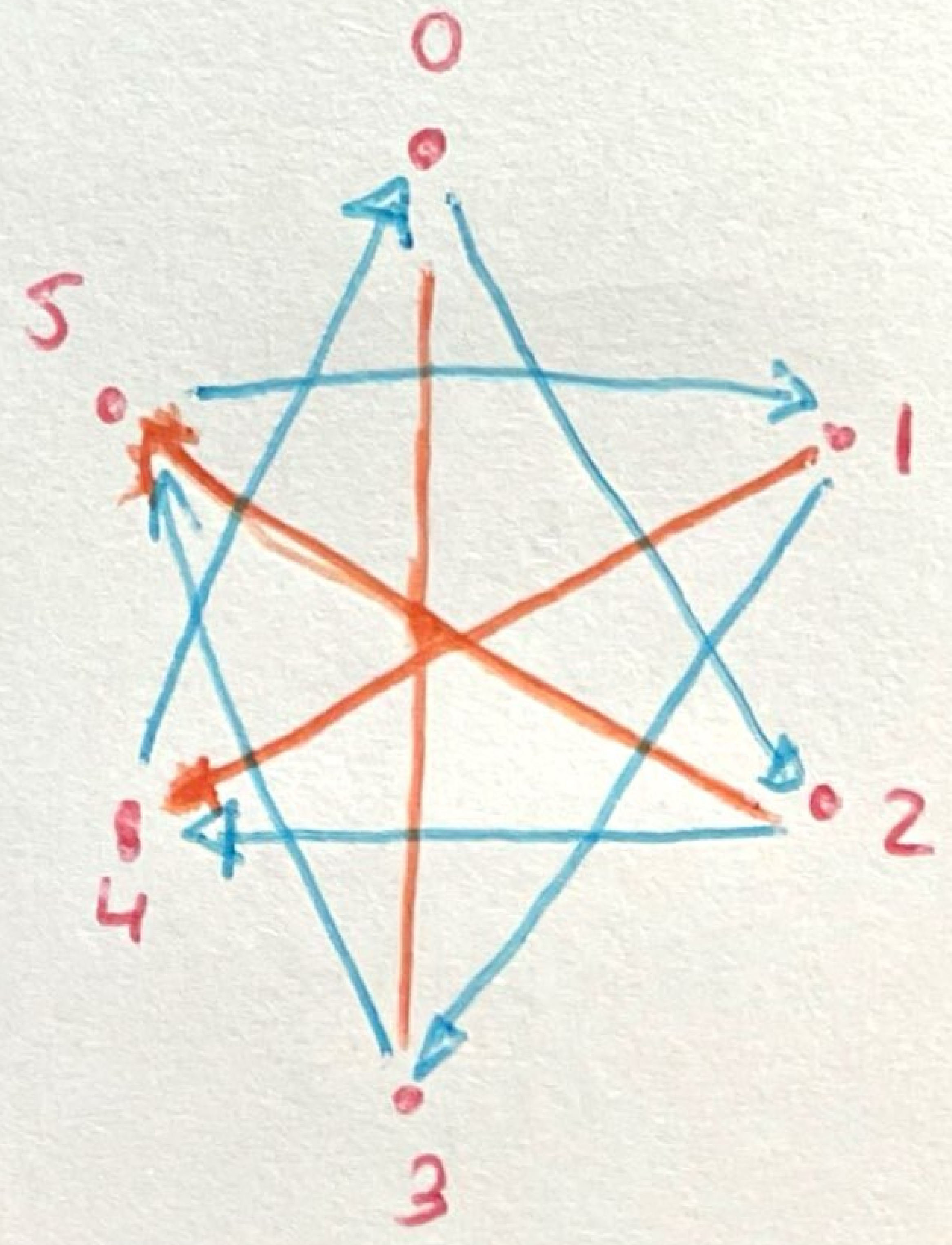
Example: a) $\mathbb{Z}_6$. and $S=\{1\}$.



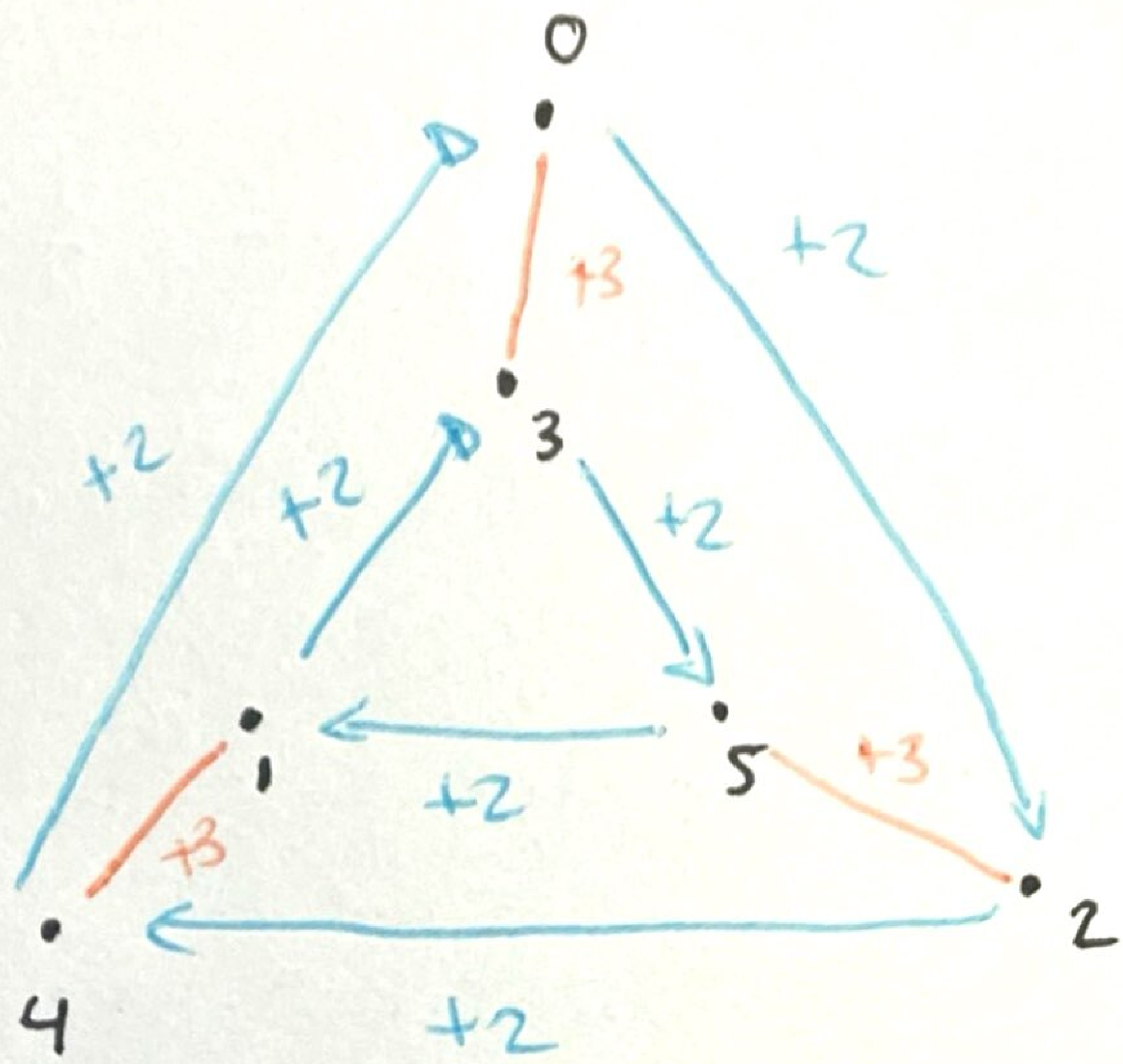
b) $\mathbb{Z}_6$ and $S=\{2,3\}$

2 = blue

3 = orange



re-organazing dots.



The point is that we reduce the grap into a graph representation and some properties of the group corespond to properties of the graph.

it gives a one to one correspondance

we have described this direction.

Cayley graps ← - - → Finite groups

↳ A graph satisfying the 4 propenties on the right.

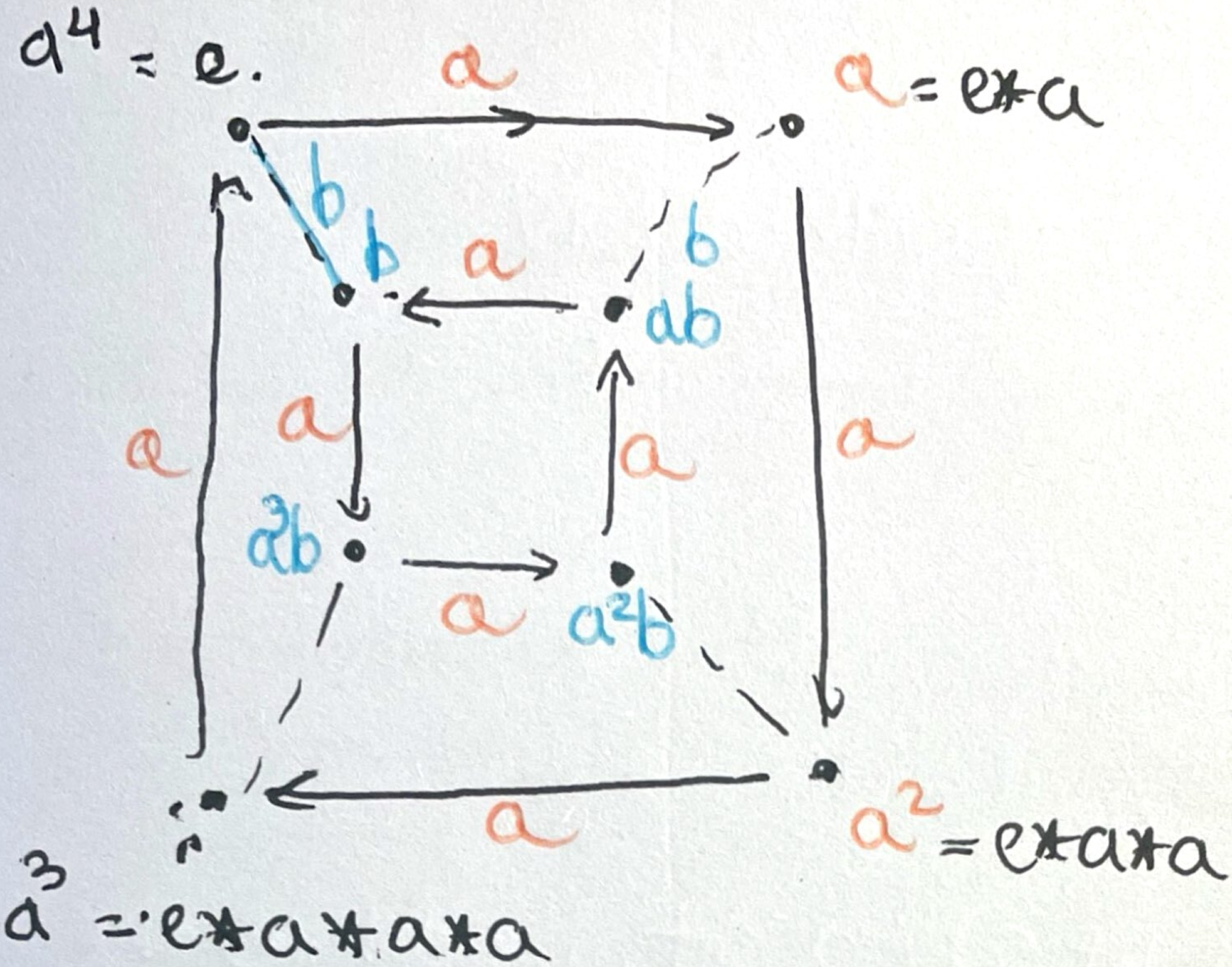
Property of the Graph	Property of the group.
1) the graph is connected for every g, h there is a path from g to h.	1) Every equation has a solution $gx = h$
2) At most one arc goes from a vertex g to a vertex h.	2) The solution to the equation is unique $gx = h$.
3) Each vertex g has exactly one arc of each type starting at g and one of of each type ending at g.	3) For $g \in G$ each generator b can compute $\underbrace{gb}_{\text{to go in / out}}$ and $\underbrace{(g\bar{b})}_{\text{arriving.}} b = g$.
4) If two different sequences of arc types starting from a vertex g lead to the same vertex h then those same seq of arc types starting from any vertex u will lead to the same vertex v.	4) If $gq = h$ and $gr = h$ Then $uq = ug^{-1}h = vr$.

How to construct the group from the graph?

The vertices ⇒ we can see them as the elements of G.

each arc says which element we are computing.
↳ generator.

Example:



Lecture 11: Permutations, Cosets and Direct Products.

GROUPS OF PERMUTATIONS

Idea: groups of matrices $GL_n(\mathbb{R})$ act on vector space.

We want to consider examples of finite groups that act on finite sets.

Definition: Let S be a set, we define the group of permutations of S to be the set of all bijections from S to S, denoted $\Sigma(S)$ where the group binary operation is composition of functions.

$(\Sigma(S), \circ, Id_S)$ is a group:

a) associativity: we want to check that given bijections $f, g, h : S \to S$ then

$$f \circ (g \circ h) = (f \circ g) \circ h.$$

But this can be shown by saying that they coincide ~~as~~ on the value of each $s \in S$.

i.e. For $s \in S$

$$(f\circ g)\circ h(s) \underset{\substack{\uparrow \\ \text{def} \\ \text{of } (f\circ g)\circ h}}{=} (f\circ g)(h(s)) \underset{\substack{\uparrow \\ \text{def of} \\ f\circ g.}}{=} f(g(h(s)))$$

and

$$f\circ(g\circ h)(s) \underset{\substack{\uparrow \\ \text{def of} \\ f\circ(g\circ h)}}{=} f(g\circ h(s)) \underset{\substack{\uparrow \\ \text{def} \\ g\circ h}}{=} f(g(h(s)))$$

since $\forall s \in S$ they agree then they are the same function.

~~inverses: given~~

• existence of identity: $\forall f \in \Sigma(S)$

$f \circ Id_S = Id_S \circ f = f$.

since $\forall s \in S$ $\quad f\circ Id_S(s) = f(Id_S(s)) = f(s)$

• inverses: this was HWK1!

You proved that given $f: S \to S$ a bijection there is $g: S \to S$ s.t. $f\circ g = g\circ f = Id_S$

~~Let ne~~

RMK: Let $n \in \mathbb{N}$ we write $Sym\, n := \Sigma(\{1,2,\dots,n\})$. If S is any set of cardinality n then $\Sigma(S) \cong Sym\, n$.

Let's look how this looks like:

Say $S = \{1, 2, 3, 4, 5\}$.

What is a bijection of S $\Rightarrow$ it is going to look like a function moving all S around. (in a one to one correspondance)

$1 \to 4$

x $2 \to 2$

x $3 \to 5$

x $4 \to 3$

x $5 \to 1$

1 2 3 4 5

4 2 5 3 1

<u>Remk:</u> A permutation of a set S is a bijection from S to S.

Definition 2 Let $(G,*,e)$ be a group and S be a set. By a group action of $(G,*,e)$ on S we mean a map

$\mu : G \times S \to S$ such that

1. $\forall x,y \in G$ and $s \in S$

$$\mu(x*y,s) = \mu(x,\mu(y,s))$$

2. $\mu(e,s) = s$. $\forall s \in S$.

Examples of group actions:

Let S be a set then $\underbrace{(\Sigma(S),\circ,Id_S)}_{G}$ acts on S.

Let $\mu : G \times S \to S$ ← it is a natural action.

$(f,s) \to f(s)$

Let's check the properties are satisfied.

1) take $f,g \in \Sigma(S)$ and $s \in S$

then $\mu(f\circ g,s) = f\circ g(s) = f(g(s))$

while $\mu(f,\mu(g,s)) = f(\mu(g,s)) = f(g(s))$

2) $\mu(Id_S,s) = Id_S(s) = s$.

[This is called a left regular representation of G.

Another example:

$GL_n(\mathbb{R})$ acts on $\mathbb{R}^2$ via which map.

- $\mu : GL_n(\mathbb{R}) \times \mathbb{R}^2 \longrightarrow \mathbb{R}^2$

$$\left(A, \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) \longrightarrow A \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

let's check thing function:

$\boxed{\text{Axiom1}}$: let $A, B \in GL_n(\mathbb{R})$

then given A, B matrices we have and $\vec{x} \in \mathbb{R}^2$

- $\mu(A \cdot B, \vec{x}) = (A \cdot B) \cdot \vec{x}$
- $\mu(A, \underbrace{\mu(B, \vec{x})}_{B\vec{x}}) = A \cdot (B\vec{x}) = (A \cdot B)\vec{x}$.

For axiom2: $\mu(I, \vec{x}) = I\vec{x} = \vec{x}$.

~~Example 3: You can~~ consider G acting on itself, when G is a group.

$\cancel{\mu : G \times G \longrightarrow G}$ ~~is an action of G on S = G.~~

$\cancel{(g, h) \longrightarrow g * h}$

$\cancel{(h, g) \longrightarrow h^{-1} * g * h}$.

Axioms: Consider G acting on itself.

$$G \times G \longrightarrow G$$
$$(h, g) \longrightarrow h * g * h^{-1}$$

Axiom 1: take $h_1, h_2 \in G$ and $g \in G$

we want $\mu(h_1 * h_2, g) = \mu(h_1, \mu(h_2, g))$

$$\mu(h_1 * h_2, g) = (h_1 * h_2) * g * (h_1 * h_2)^{-1}$$

$$\mu(h_2, g) = h_2 g h_2^{-1}$$

$$\mu(h_1, \mu(h_2, g)) = h_1 \mu(h_2, g) h_1^{-1}$$

$$= h_1 * (h_2 * g * h_2^{-1}) * h_1^{-1}$$

$$\underset{\text{ass.}}{=} (h_1 * h_2) * g * \underbrace{(h_2^{-1} * h_1^{-1})}_{= (h_1 * h_2)^{-1}}$$

This has a name! it is called the conjugation action.

Most important example of an action of G into itself:

$$\mu : G \times G \longrightarrow G$$
$$(g, h) \longrightarrow g * h.$$

← the natural action.

Let $\psi : G \times S \to S$ an action from a group G ~~into itself~~ on a set S.

Then if we fix an element $g \in G$ It naturally gives rise to a map

$$\psi_g : S \to S$$
$$s \to \psi(g, s)$$

Note that if ψ is an action then axiom 1 guarantees that $\forall g, h \in G$.

$$\boxed{\psi_{g*h} = \psi_g \circ \psi_h.} \quad *$$

why?

$$\underbrace{\psi(g*h, s)}_{\psi_{g*h}(s)} = \underbrace{\psi(g, \underbrace{\psi(h, s)}_{\psi_h(s)})}_{\psi_g(\psi_h(s))}$$

And property (2) makes sure that

$$\boxed{\psi_e = Id_S.} \quad *2.$$

~~then:~~ ~~Proposition 1: For any~~

Proposition 1: Let $\psi: G \times S \to S$ an action an $g \in G$ fixed let

$$\begin{aligned}\psi_g : S &\longrightarrow S \\ s &\longrightarrow \psi(g,s)\end{aligned}$$

then ψ_g is a bijection.

Proof:

~~injective: Assume $\psi(g,s_1) = \psi(g,s_2)$~~

~~then ψ_g~~,

Note that ψ_g has an inverse which is $\psi_{g^{-1}}$, since

$$\begin{array}{c}\psi_{g * g^{-1}} = \psi_e = Id_S \\ \shortparallel \\ \psi_g \circ \psi_{g^{-1}}\end{array}$$

similarly $\psi_{g^{-1}} \circ \psi_g = \psi_e = Id$

thus ψ_g is a bijection.

In particular we have an embedding

$$\begin{aligned}(G,*) &\longrightarrow (\Sigma(S), \circ, Id_S) \\ g &\longrightarrow \psi_g\end{aligned}$$

And it is an homomorphism.

$*$ says exactly this

Definition: An action G on S is faithful

if $\varphi : G \longrightarrow \cancel{\Sigma(G}\ \Sigma(S)$

$$g \longmapsto \varphi_g$$

is injective.

In particular if φ is injective then G is isomorphic to a subgroup of $(\Sigma(S), \circ, Id_S)$

Cayley's theorem:

Let $(G, *, e)$ be a group, then G is isomorphic to a subgroup of $\Sigma(G)$.

In particular if $|G| = n$ then G is isomorphic to a subgroup of Sym_n.

Proof: Consider the action of G into itself given by $\mu : G \times G \longrightarrow G$

$$(g, s) \longmapsto g * s.$$

(multiplying by g on the left)

then we have that for each $g \in G$ the map $\varphi_g : G \longrightarrow G$ is a bijection.

thus we can consider the embedding

$$\varphi : (G, *, e) \longrightarrow (\Sigma(G), \circ, Id_G)$$

$$g \longrightarrow \varphi_g$$

it is well defined and moreover it is an homomorphism (it is a particular case of the previous case since $G = S$).

Then we just need to explain that the action of G into itself is faithful, i.e. the map φ is injective.

so let $g_1, g_2 \in G$ and assume $\varphi_{g_1} = \varphi_{g_2}$

we want to show that $g_1 = g_2$.

since $\varphi_{g_1} = \varphi_{g_2}$ in particular $\varphi_{g_1}(e) = \varphi_{g_2}(e)$

$\Rightarrow g_1 * e = g_2 * e \Rightarrow g_1 = g_2$ as required.

More on permutation groups

Baby example: $A = \{1,2,3,4,5\}$

$\sigma \in \Sigma(A)$

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix}$$ thus $\sigma(1) = 4$

$\sigma(2) = 2$.

$$\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}$$

Then how does $\sigma \circ \tau$ look like?
Follow first τ then apply σ.

$$\sigma \circ \tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 1 & 3 & 2 & 4 \end{pmatrix}$$

$1 \to 3 \to 5$

$2 \to 5 \to 1$

$3 \to 4 \to 3$

$4 \to 2 \to 2$

$5 \to 1 \to 4$

2 examples on the finite world:

$S_3 = \cancel{6}\; 3! = 3\cdot 2\cdot 1 = 6.$

<u>We can list them:</u>

$\rho_0 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ ← identity $\qquad \mu_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$

$\rho_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \qquad \mu_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$

$\rho_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \qquad \mu_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$

<u>table of operations:</u>

	ρ_0	ρ_1	ρ_2	$\cancel{\rho_0}\,\mu_1$	μ_2	μ_3
ρ_0	ρ_0	ρ_1	ρ_2	μ_1	μ_2	μ_3
ρ_1	ρ_1	ρ_2	ρ_0	μ_3	μ_1	μ_2
ρ_2	ρ_2	ρ_0	ρ_1	μ_2	μ_3	μ_1
μ_1	μ_1	μ_2	μ_3	ρ_0	ρ_1	ρ_2
μ_2	μ_2	μ_3	μ_1	ρ_2	ρ_0	ρ_1
μ_3	μ_3	μ_1 $\cancel{\mu_3}$	μ_2	$\cancel{\mu_2}\,\rho_1$	ρ_2	ρ_0.

Note that is NOT ABELIAN.

There is a natural correspondance between the elements of S_3 and the way in which 2 copies of an equilateral triangle with vertices 1, 2 and 3 can be placed.

S_3 is also called the group D_3 of symmetries of an equilateral triangle.

$\rho_i \to$ are rotations

$\mu_i \to$ mirror images of bisector angles.

The n-th Dihedral group D_n is the group of symmetries of the regular n-gon.